

Trend Following, Leveraged Re-Entry, and Volatility Decay

S&P 500 total return. One signal only: the 200-day moving average (the daily equivalent of Faber's 10-month rule). Tested as far back as the data allows.

Data. Total return throughout. Monthly S&P 500 total return back to 1901 (Shiller) for the Faber replication; a daily total return series from 1928 for everything else — real $\wedge SP500TR$ from 1988, and before that $\wedge GSFC$ price plus the Shiller dividend yield (this reconstruction tracks the real series with 0.50%/yr error and 0.9996 correlation over 1988–2026). Cash earns the 13-week T-bill ($\wedge IRX$; a 3.5% constant before 1960). The trend signal is lagged one day, so nothing uses information we could not have had.

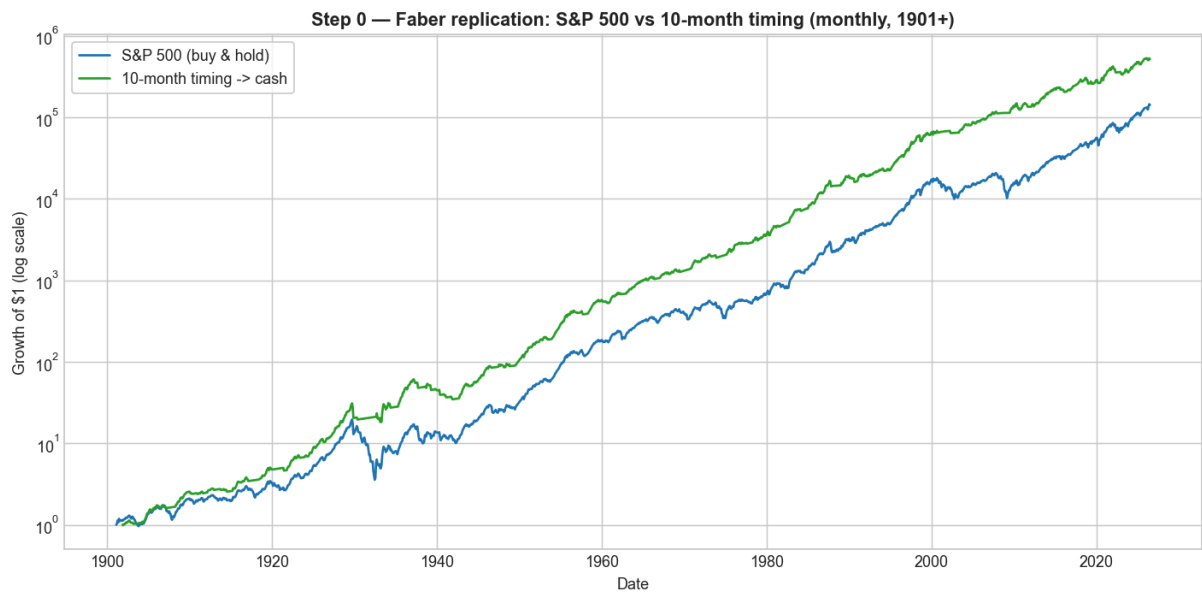
1. Buy & hold vs the Faber moving-average rule

The rule: hold the S&P 500 while it is **above** its moving average; move to **cash** while it is **below**. Faber uses the 10-month SMA on monthly data; we use that, then its daily twin (the 200-day SMA).

Monthly, 1901–2026 (Faber's exact setup):

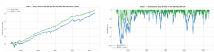
	S&P 500 buy & hold	10-month timing → cash
CAGR	9.95%	11.24%
Volatility	15.4%	10.8%
Sharpe	0.44	0.69
Max drawdown	–81.8%	–43.0%

(Faber's published drawdowns are –83.66% → –42.24%; we reproduce them to within a point.)



Daily, 200-day SMA, 1928–2026:

	Buy & Hold 1x	MA200 → Cash
CAGR	10.14%	11.29%
Volatility	18.9%	12.6%
Sharpe	0.40	0.60
Sortino	0.56	0.84
Max drawdown	–83.9%	–46.2%
Calmar	0.12	0.24



The moving-average rule keeps essentially all of the return while cutting volatility by a third and halving the worst drawdown, so its Sharpe, Sortino and Calmar are all much higher. The 200-day MA adds clear risk-adjusted value.

2. Leverage returns

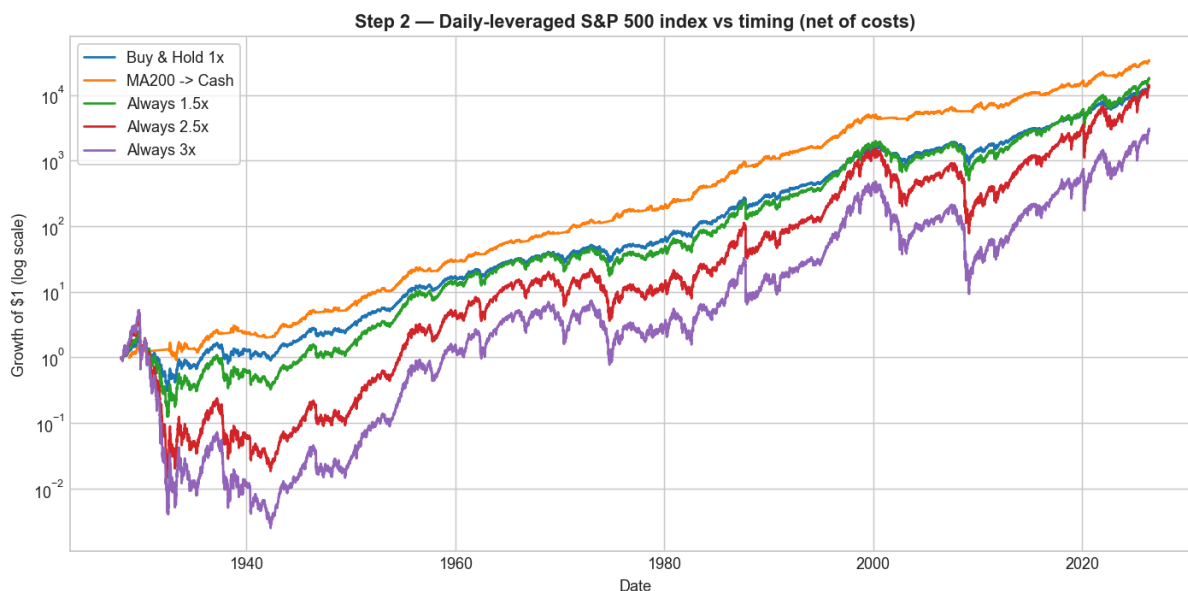
A daily **Lx leveraged** return is simply that day's S&P 500 total return multiplied by L, and then compounded day by day:

$$\text{leveraged_return}[t] = L \times \text{sp500_return}[t] \quad (\text{before fees / financing})$$

For example, the 2x series is just the daily S&P total return $\times 2$, compounded. Borrowed money costs the financing rate, and leveraged funds charge a fee ($\sim 0.9\%$ /yr); both are included below.

Holding **constant** daily leverage on the index, net of costs:

	CAGR	Grew \$1 to
1x (buy & hold)	10.14%	\$13,021
Always 1.5x	10.46%	\$17,378
Always 2.5x	10.15%	\$13,110
Always 3x	8.43%	\$2,803



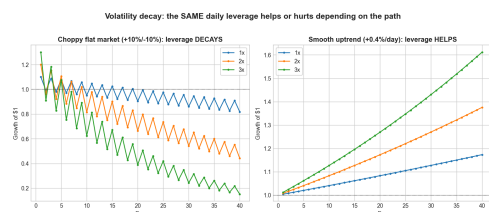
Constant leverage barely helps at 1.5x and *loses ground* by 3x — across a full century (including 1929 and 2008), constant 3x ends below plain buy & hold and far below the moving-average rule.

3. Volatility decay, and buying leverage at the lows

Volatility decay. A +10% day followed by a -10% day:

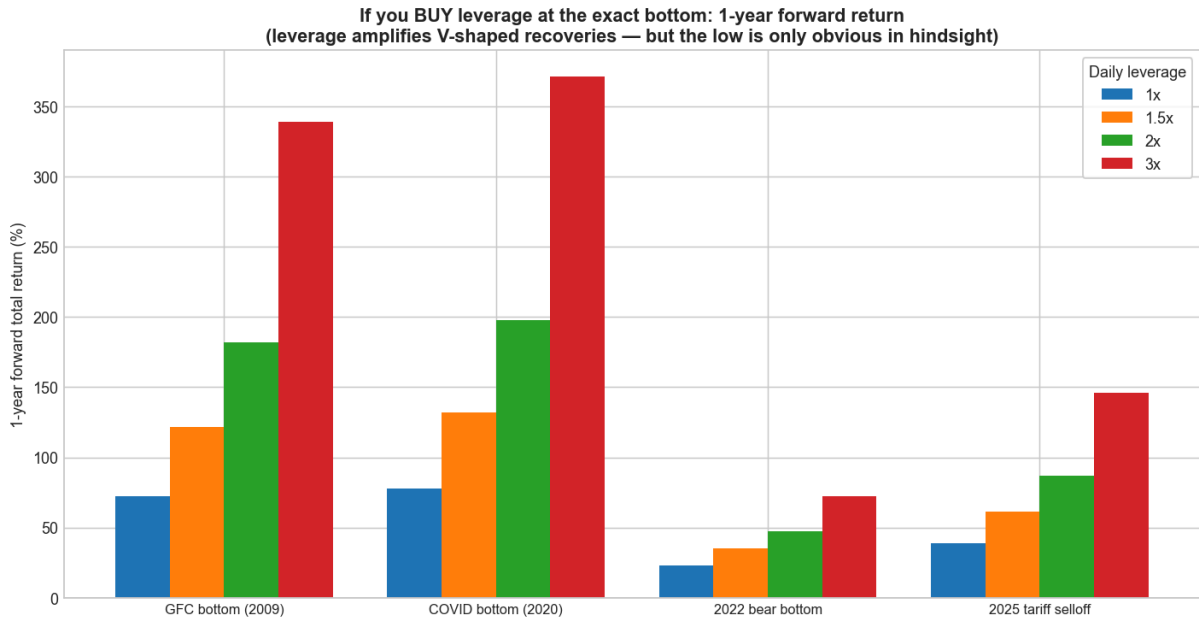
	1x	2x	3x
Two-day return	-1.0%	-4.0%	-9.0%

The market round-trips to roughly flat, but leverage loses — and the loss grows with the *square* of leverage. The annual penalty is $\approx \frac{1}{2} \cdot L^2 \cdot \sigma^2$: at 20% volatility, $\sim 2\%$ /yr for 1x, **8%/yr for 2x**, **18%/yr for 3x**.



But decay only bites in choppy/falling markets. In a one-directional rally — like the rebound off a market bottom — leverage amplifies the gain. Forward **1-year** total return if you had bought at the exact low:

Bottom	1×	1.5×	2×	3×
GFC (2009-03-09)	+72%	+122%	+182%	+339%
COVID (2020-03-23)	+78%	+132%	+198%	+372%
2022 (2022-10-12)	+23%	+35%	+47%	+72%
2025 tariff selloff (2025-04-08)	+39%	+61%	+87%	+146%



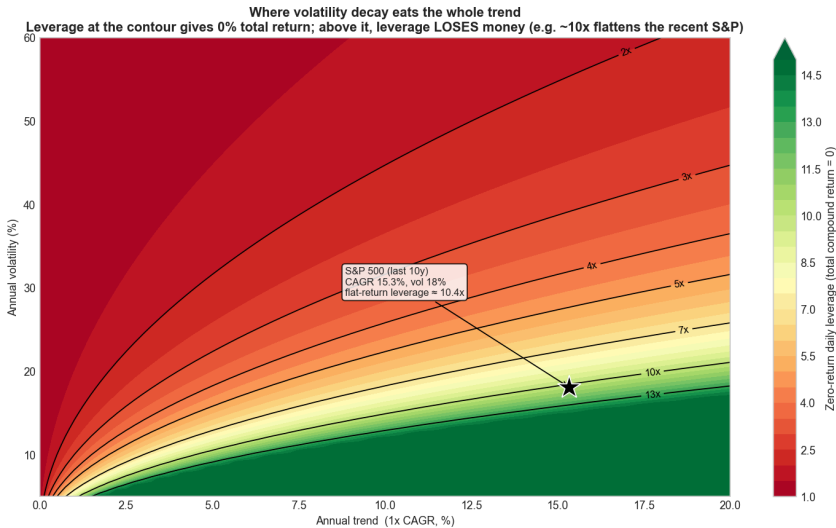
Leverage can work strongly in your favour **if you time it** — buying into the recovery off a low. (The low, of course, is only obvious in hindsight.)

4. The leverage at which the total return goes flat

For a given trend (the 1× CAGR g) and volatility σ , the compound return of $L\times$ leverage is $L \cdot \mu - \frac{1}{2} \cdot L^2 \cdot \sigma^2$. Setting it to zero gives the leverage at which volatility decay exactly cancels the trend, so the **total return is flat (0%)**:

$$L_{\text{zero}} = 2 \cdot g / \sigma^2 + 1$$

Below it, leverage still grows; above it, leverage **loses money**. The map below plots L_{zero} for every combination of trend and volatility. For the S&P over the **last 10 years** (CAGR 15.3%, volatility 18.1%) the flat point is $\approx 10\times$ — so a hypothetical "10× S&P" would have gone essentially nowhere despite a strong decade, while higher still would have bled toward zero.

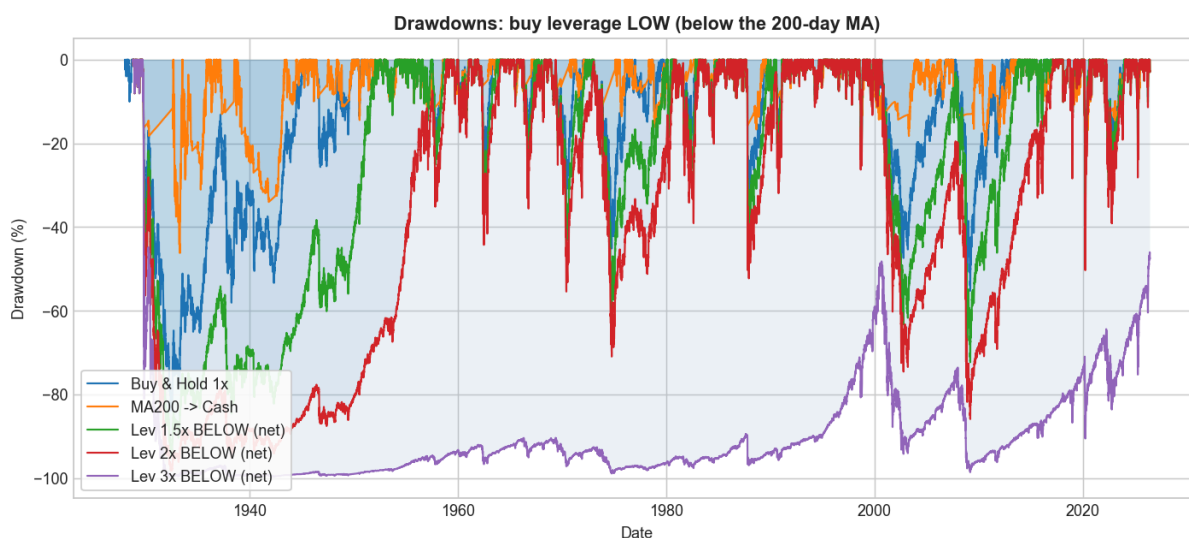
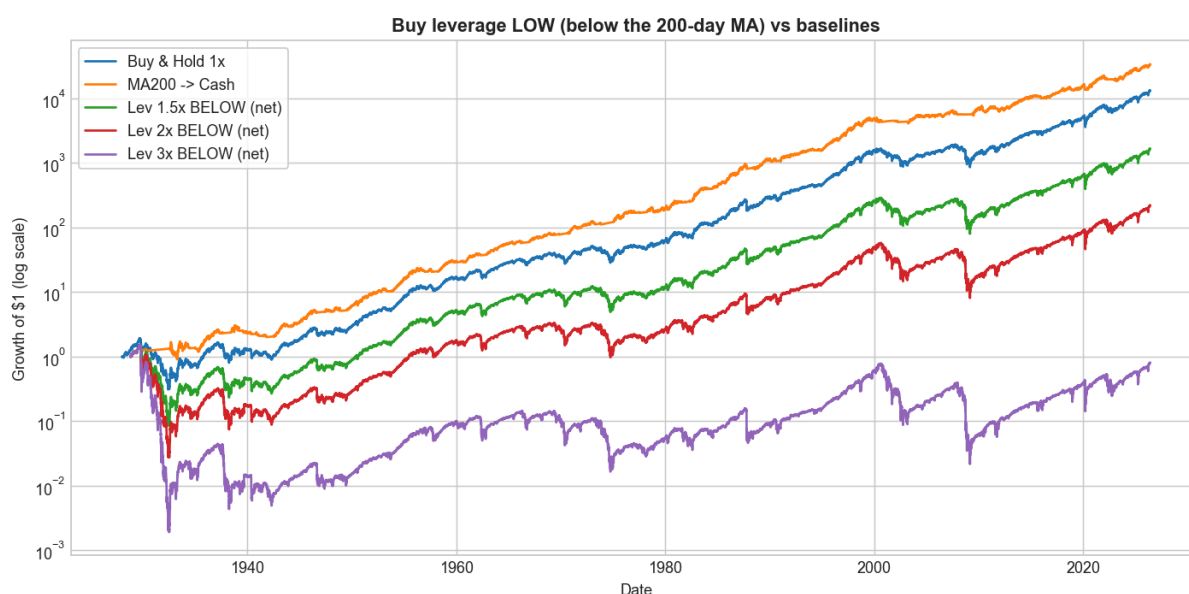


(For reference, the leverage that merely *ties* 1× buy & hold is lower — $\approx 3.1\times$ for the S&P — and the growth-maximising level is $\approx 2\times$; see [charts/F3_breakeven_leverage_map.png](#) and [charts/F3_optimal_leverage_curve.png](#).)

5. The switching strategy: buy leverage low, sell high

Use the 200-day MA to switch between leveraged and ordinary exposure. The intuitive version — motivated by the bought-the-low returns in §3 — is to **buy leverage when the market is low** (below the MA, "on sale") and drop back to plain 1× when it is high (above the MA). Daily total return, 1928–2026, net of costs:

Strategy	Grew \$1 to	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar
Buy & Hold 1×	\$13,021	10.1%	18.9%	0.40	0.56	−83.9%	0.12
MA200 → Cash	\$33,090	11.3%	12.6%	0.60	0.84	−46.2%	0.24
Lev 1.5× below MA	\$1,635	7.9%	24.7%	0.27	0.38	−94.3%	0.08
Lev 2× below MA	\$217	5.7%	31.0%	0.21	0.29	−98.2%	0.06
Lev 3× below MA	\$0.79	−0.2%	44.3%	0.13	0.19	−99.9%	−0.00

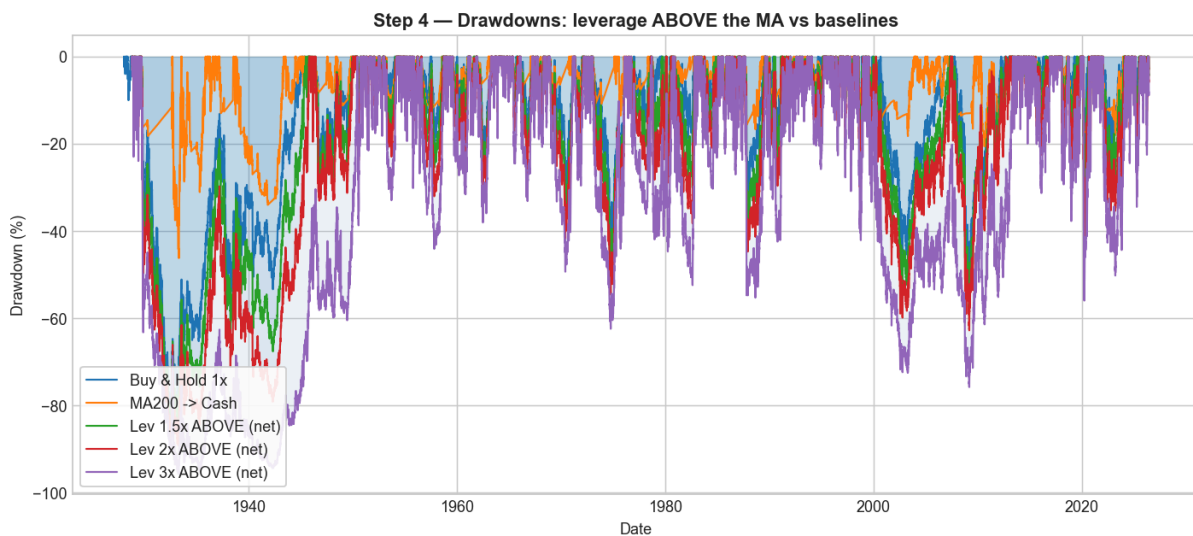
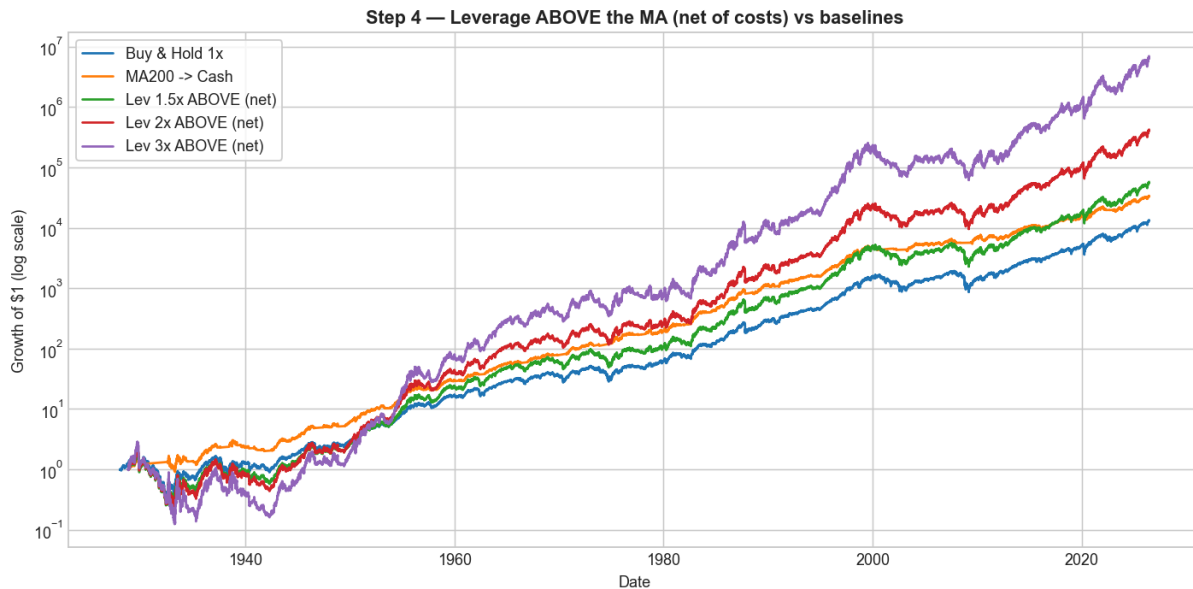


Buying leverage low gets **worse** as leverage rises — 1.5× below trails buy & hold, 2× below turns \$1 into \$217 (vs \$13,021), and 3× below actually *loses money* (\$1 → \$0.79) with a −99.9% drawdown. The reason: the 200-day MA does not buy the bottom. It flags "below trend" at the *start* of a decline, when the market is highly volatile and often still falling (1929, 2000–02, 2008) — exactly where volatility decay is worst.

6. The other way round: leverage the uptrend

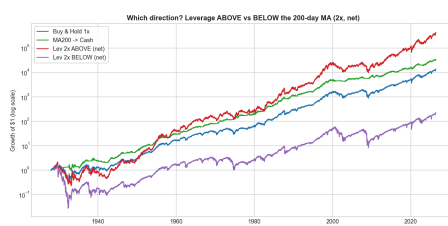
So we check the opposite switch — **leverage when the market is ABOVE the MA** (the calm, rising regime) and drop to plain 1× when it falls below. Same data, same costs:

Strategy	Grew \$1 to	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar
Buy & Hold 1×	\$13,021	10.1%	18.9%	0.40	0.56	−83.9%	0.12
MA200 → Cash	\$33,090	11.3%	12.6%	0.60	0.84	−46.2%	0.24
Lev 1.5× above MA	\$55,471	11.9%	23.6%	0.43	0.60	−85.7%	0.14
Lev 2× above MA	\$402,863	14.2%	28.9%	0.47	0.66	−89.2%	0.16
Lev 3× above MA	\$6,429,403	17.5%	40.4%	0.50	0.71	−95.7%	0.18



Leveraging the uptrend gets **better** as leverage rises, beating buy & hold on CAGR, Sharpe, Sortino and Calmar at every level — though its maximum drawdown is deeper than buy & hold (you are leveraged going *into* fast crashes), and it does not beat the plain move-to-cash rule on a risk-adjusted basis.

It is the same idea pointed in opposite directions. At 2×, the two versions differ by a factor of ~1,800× in final wealth (\$402,863 above vs \$217 below):



Educational research only — not investment advice. Reproduce with `python run_faber_leverage.py`; figures are in `charts/`, numbers in `results/`.